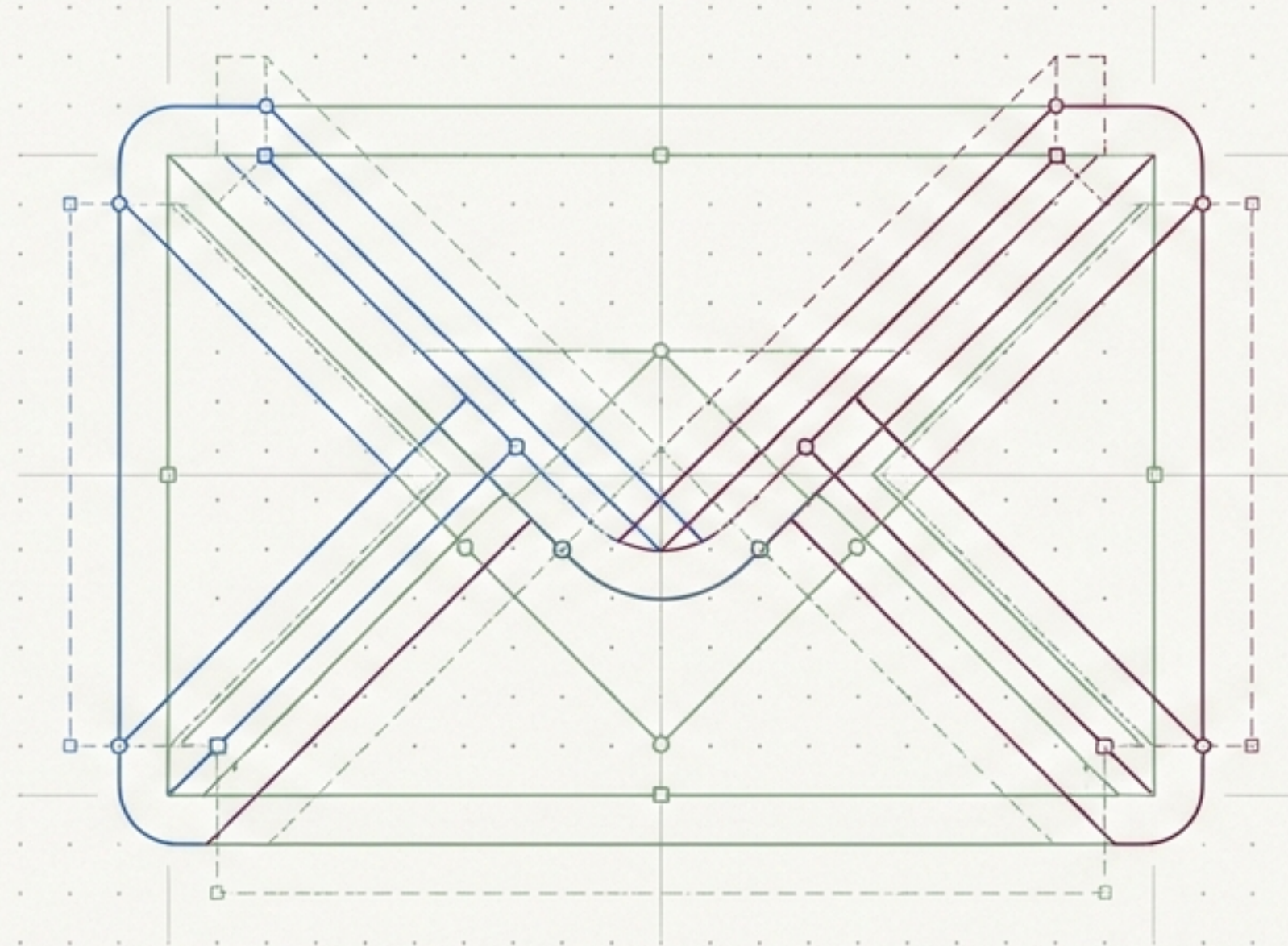




Clarion AI: The Engine Behind Autonomous Email

System Architecture and Data Flow Lifecycle

Four Pillars Power the Clarion Architecture.

Chrome Extension (MV3)

Client / Execution. Runs on Outlook Web, captures tokens, syncs emails, delivers drafts.

Web Dashboard

User Interface. React frontend at clarion-ai.app for reviewing notable emails and guiding model feedback.

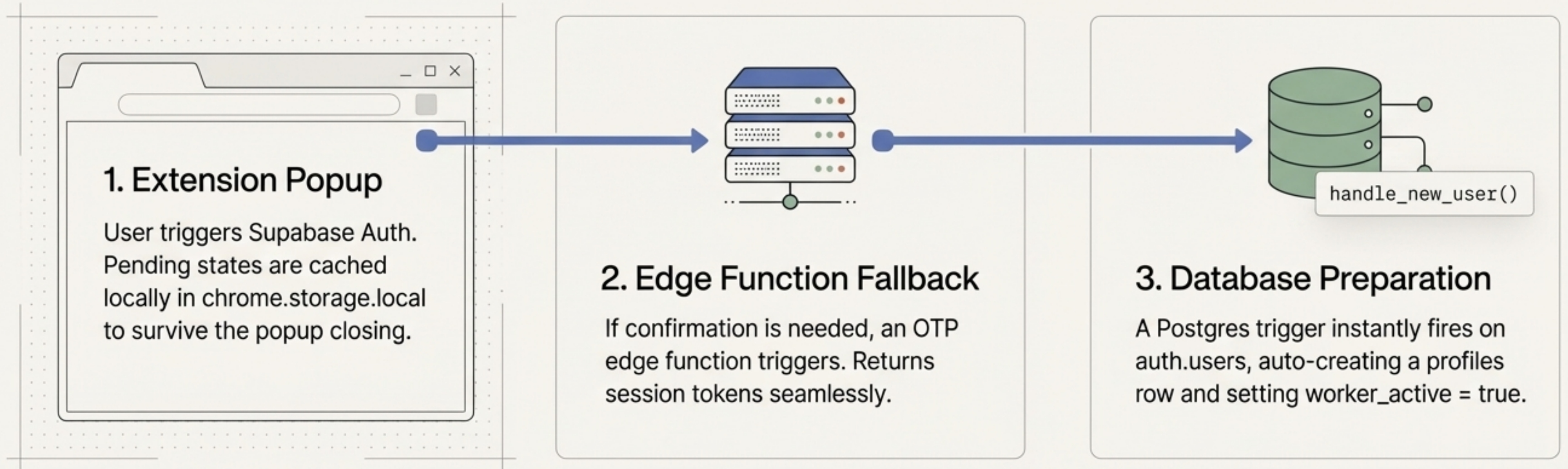
Supabase Hub

Infrastructure / Auth. Postgres DB, Auth, Realtime WebSockets, Edge Functions, Row Level Security.

Railway Worker

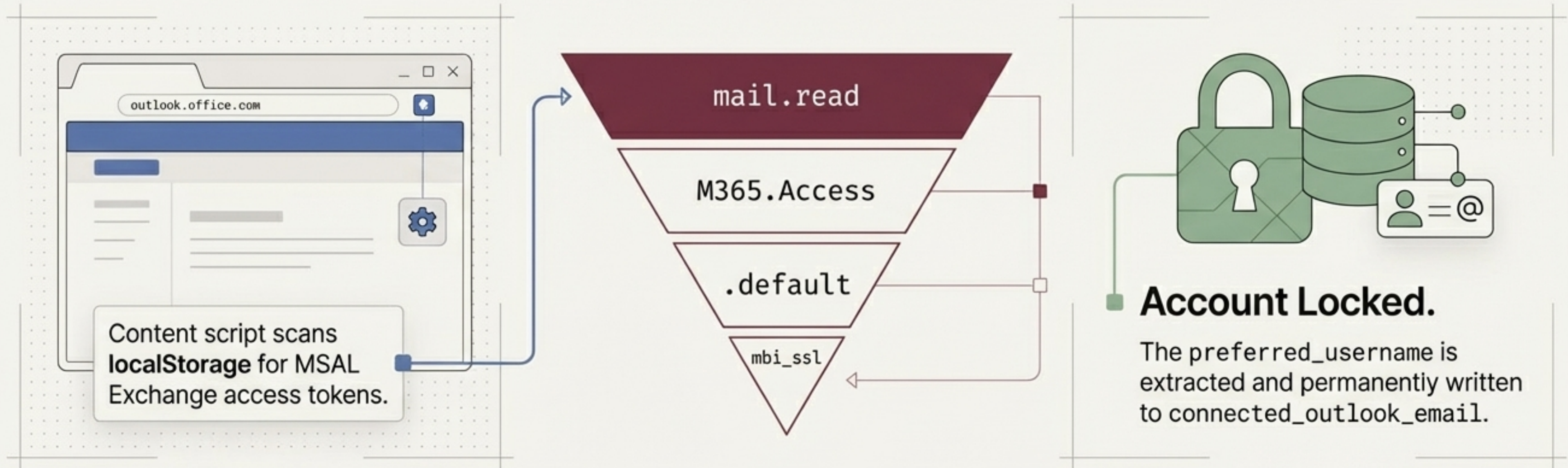
Compute / AI. Continuous Python loop handling classification and Anthropic AI batch orchestrations.

Secure Setup Operates With Zero User Friction.



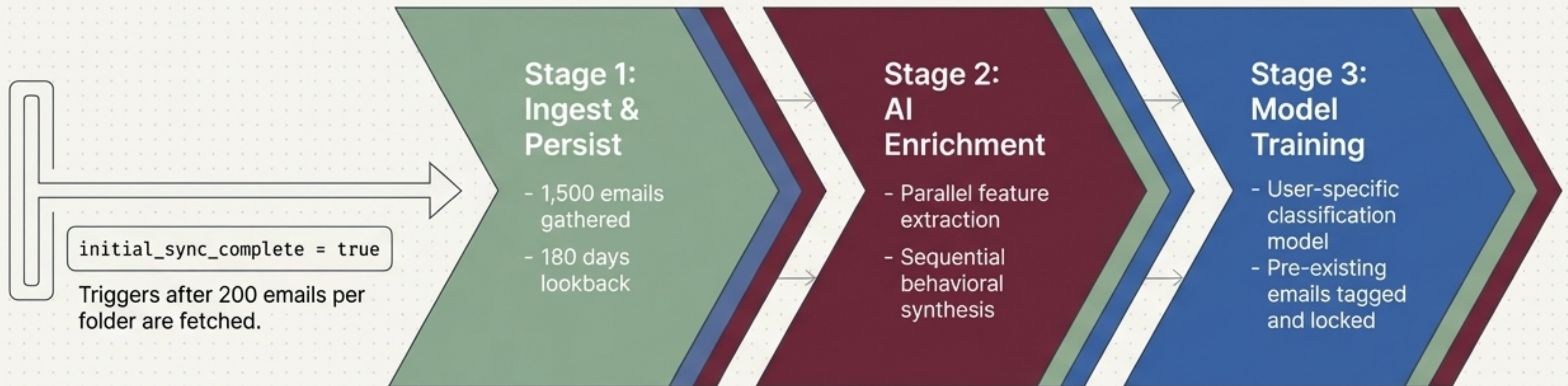
Zero-friction persistence: The user interface never stalls while database architecture is prepped in the background.

Token Management Requires No Direct Credentials.



Fail-Closed Safety: All subsequent tokens are strictly verified against the locked profile. Mismatches are rejected instantly to preserve data boundaries.

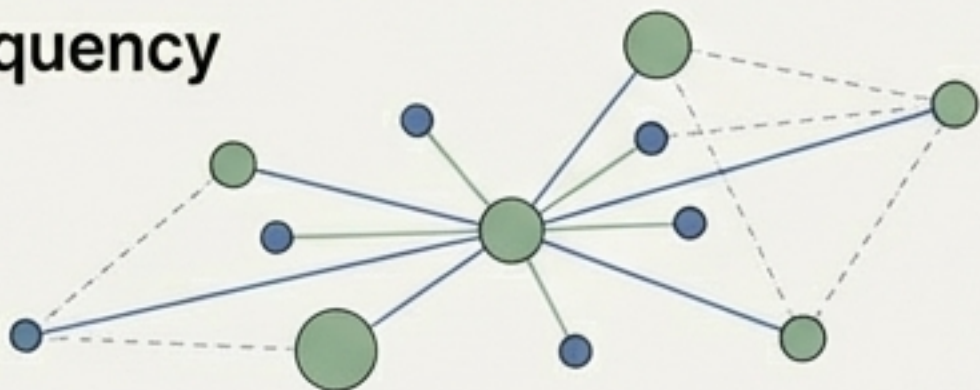
Initial Sync Triggers the Three-Stage Onboarding Engine



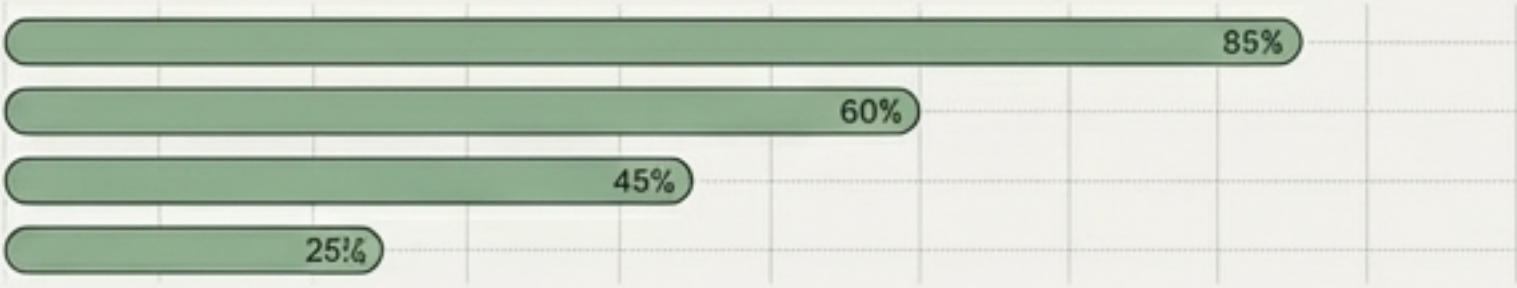
Pre-AI Statistical Analysis Maps the Social Graph.

Building the baseline: Pure Python extracts robust statistical data before any LLMs are invoked.

Contact Frequency



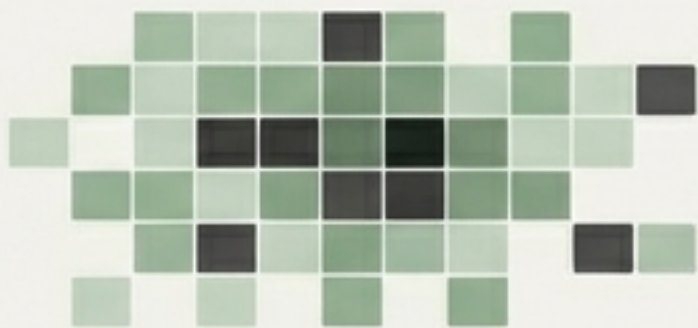
Response Rates



Thread Patterns

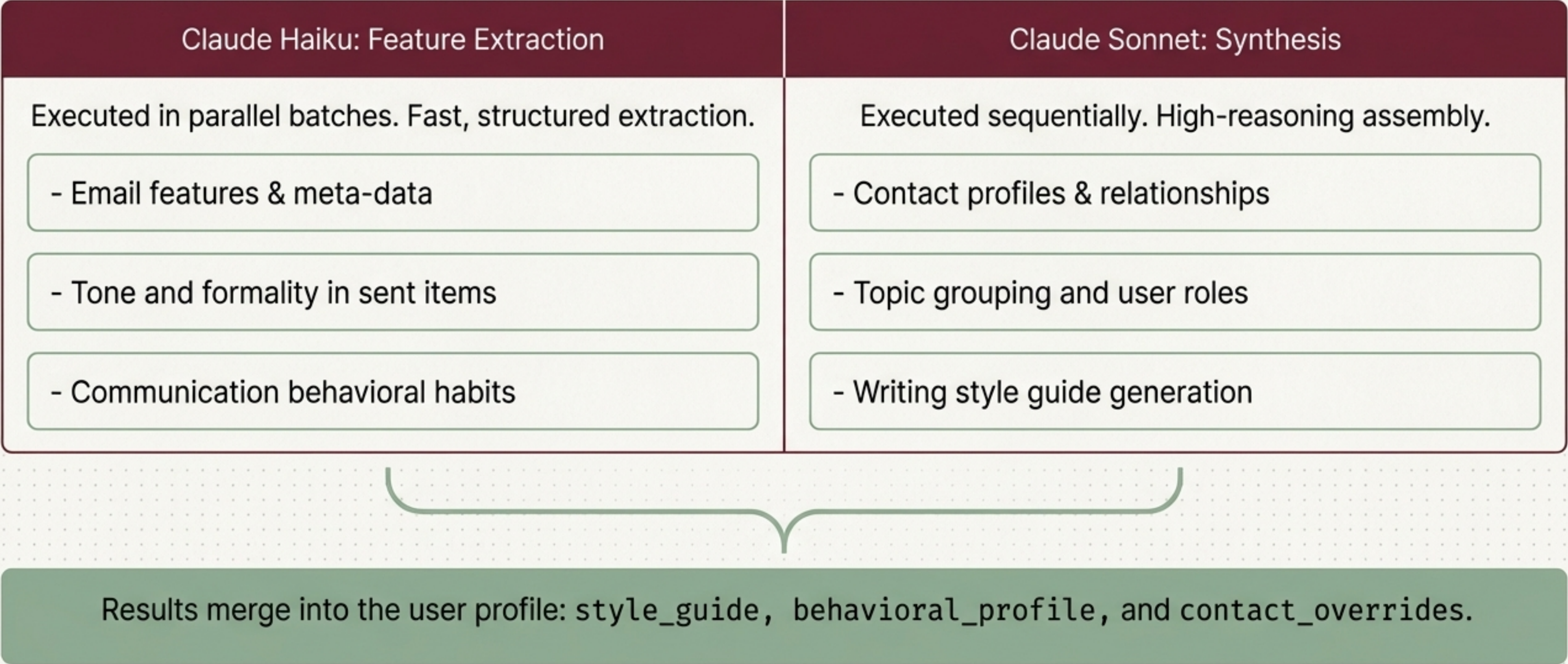


Domain Mapping

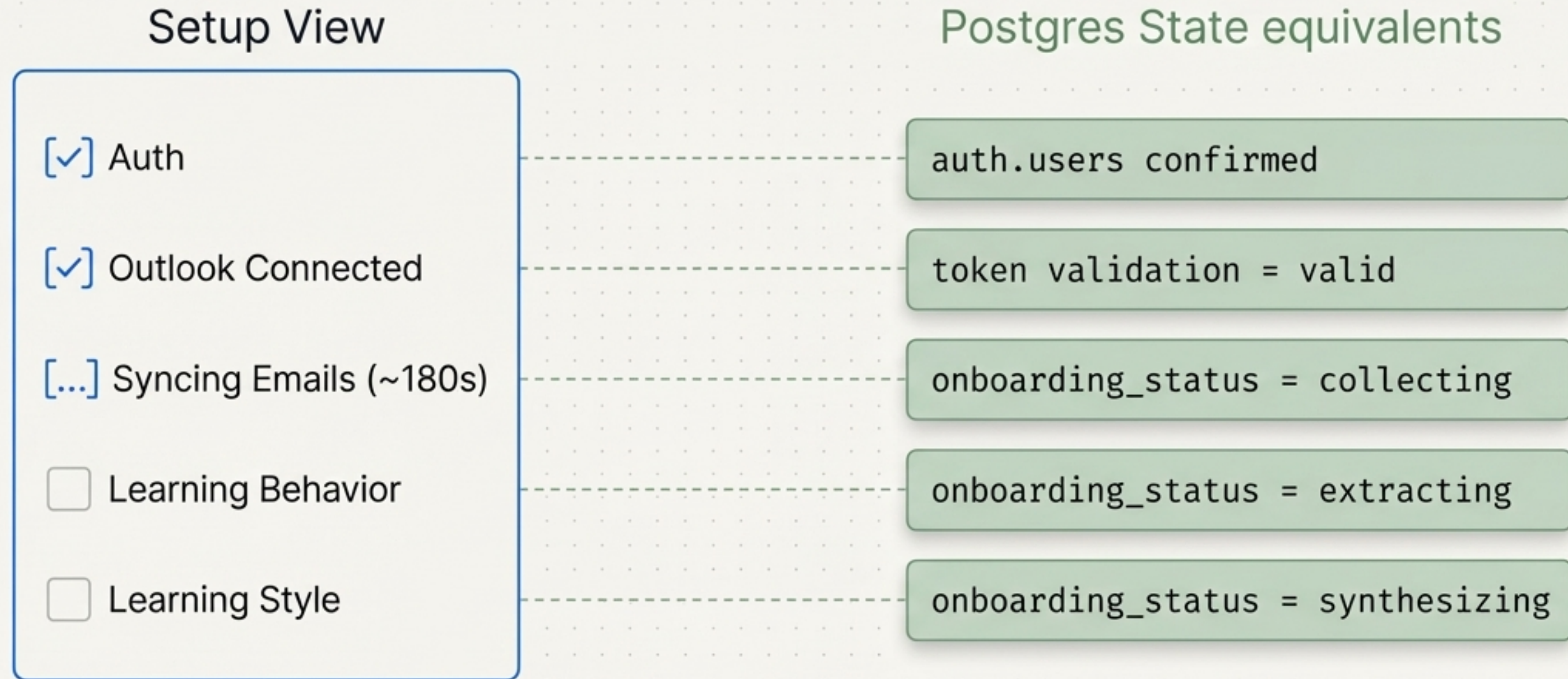


Data persisted directly to **profiles.onboarding_data**
pending → collecting → statistics → persisting

Dual-Model Architecture Balances Scale With Deep Reasoning.

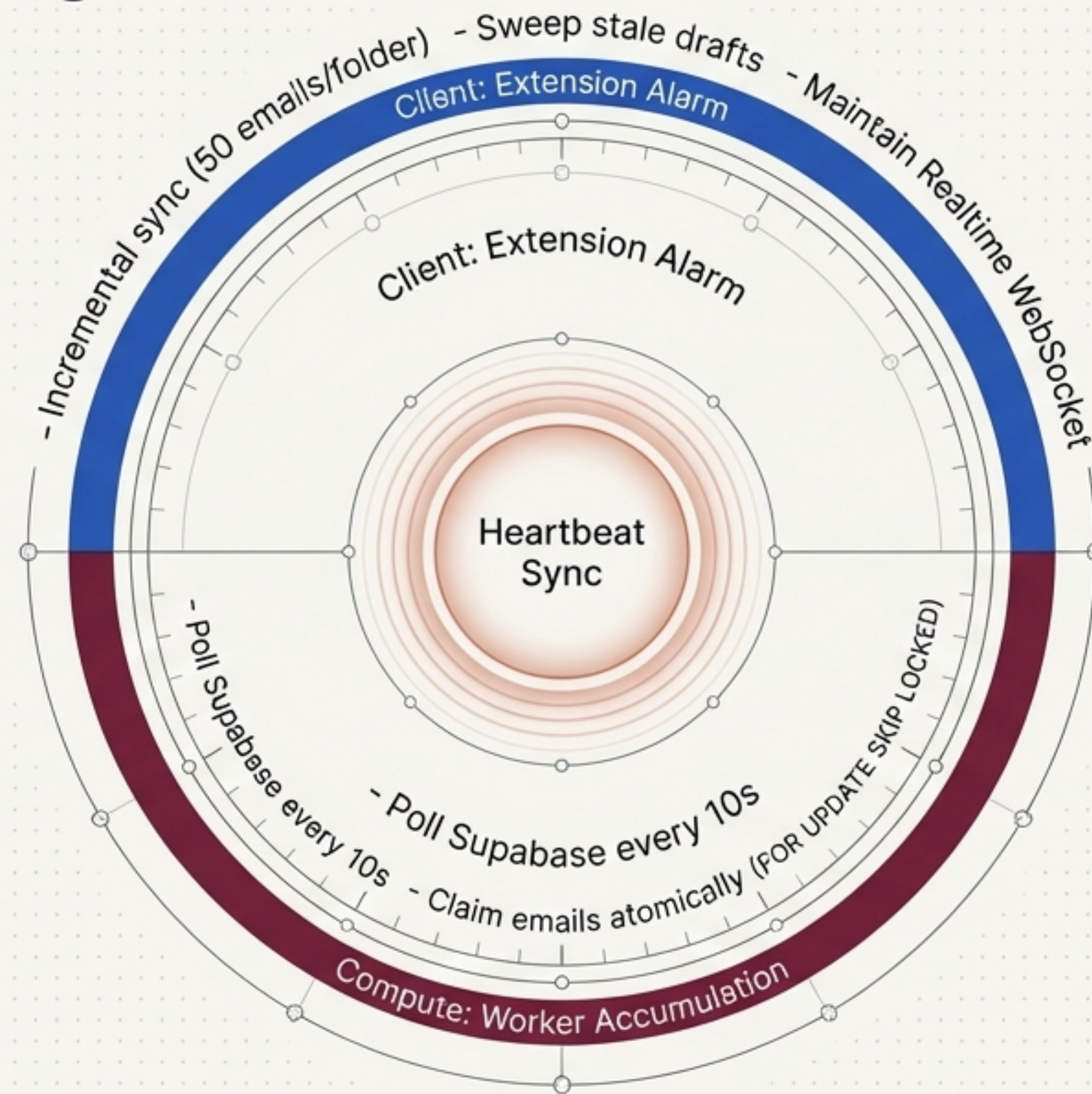


Backend Complexity Translates to Transparent User Feedback



A 5-second polling loop ensures the user is never left wondering what the engine is doing.

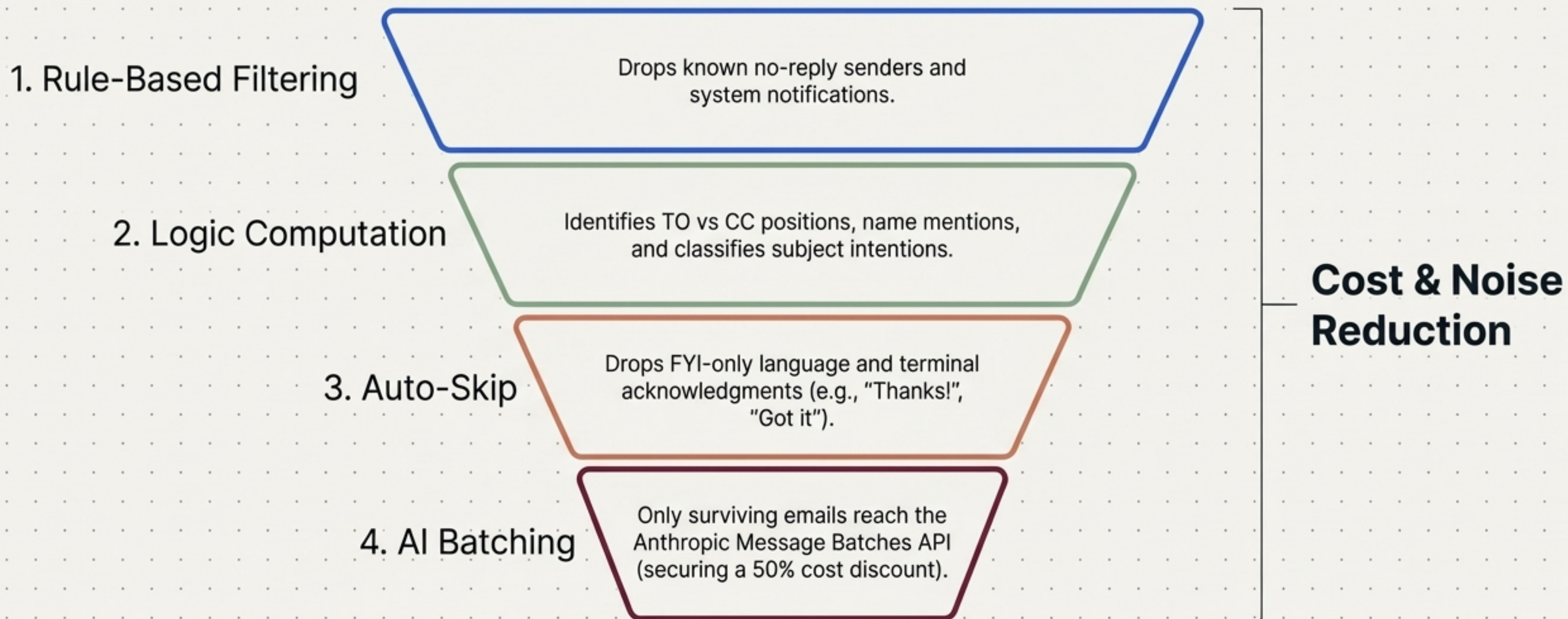
The Continuous Engine Operates on a 45-Second Loop.



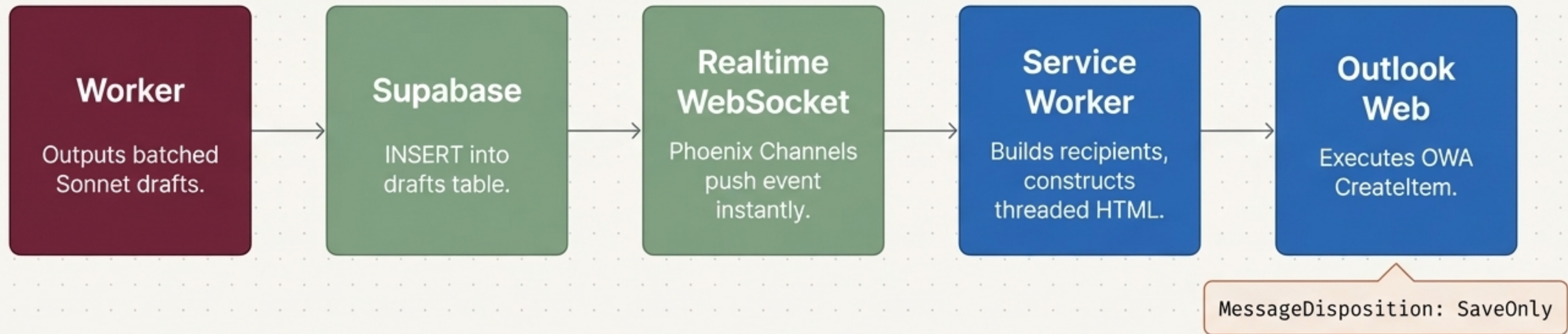
The extension sends a continuous heartbeat on every sync, keeping the active session registered with the processing worker.

Maximum Intelligence Requires Minimal Waste.

Aggressive noise filtering ensures only high-signal data incurs LLM processing costs.



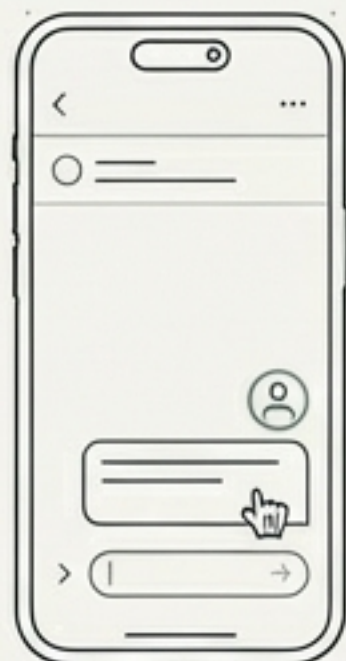
Realtime Sockets Deliver Drafts Invisibly.



Drafts are prepared entirely in the background, waiting silently in the Drafts folder for user review.

Self-Healing Mechanisms Prevent State Conflicts.

The Action (Manual Override)



User replies manually, bypassing the extension and the AI entirely.

The Correction (System Resilience)

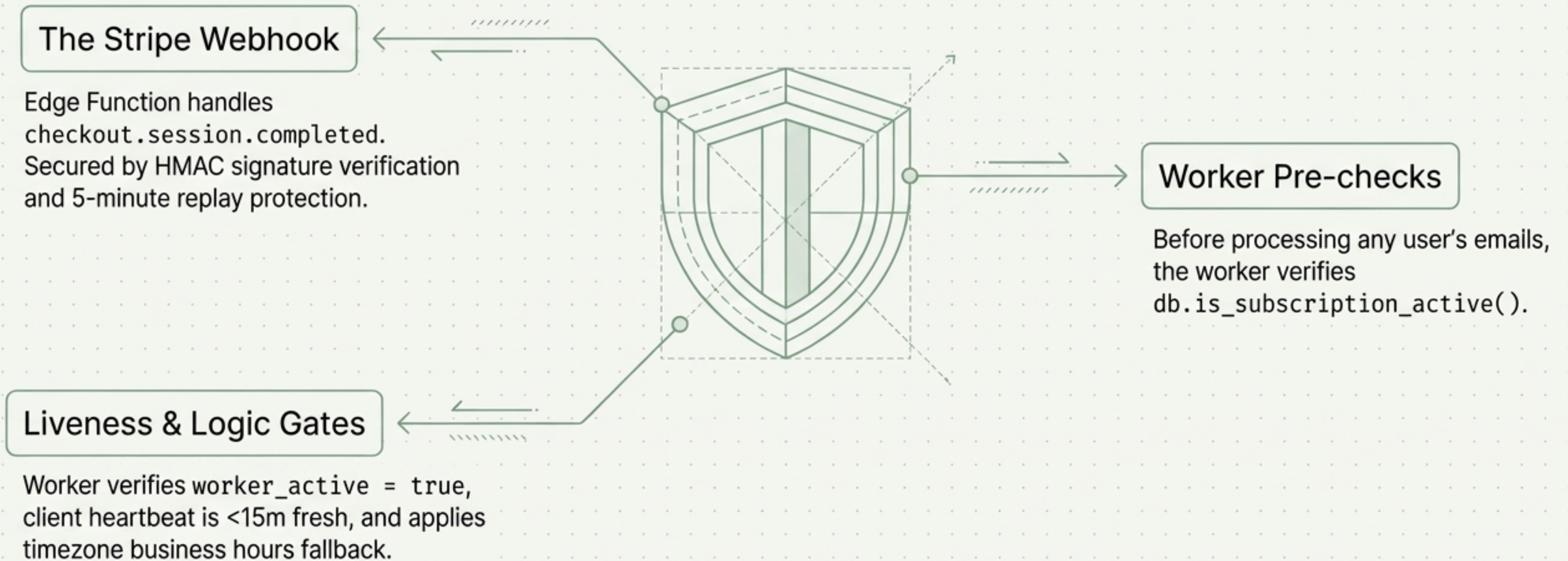
Extension 45s alarm detects the new sent item.

Fires Supabase RPC call:
`find_stale_drafts`

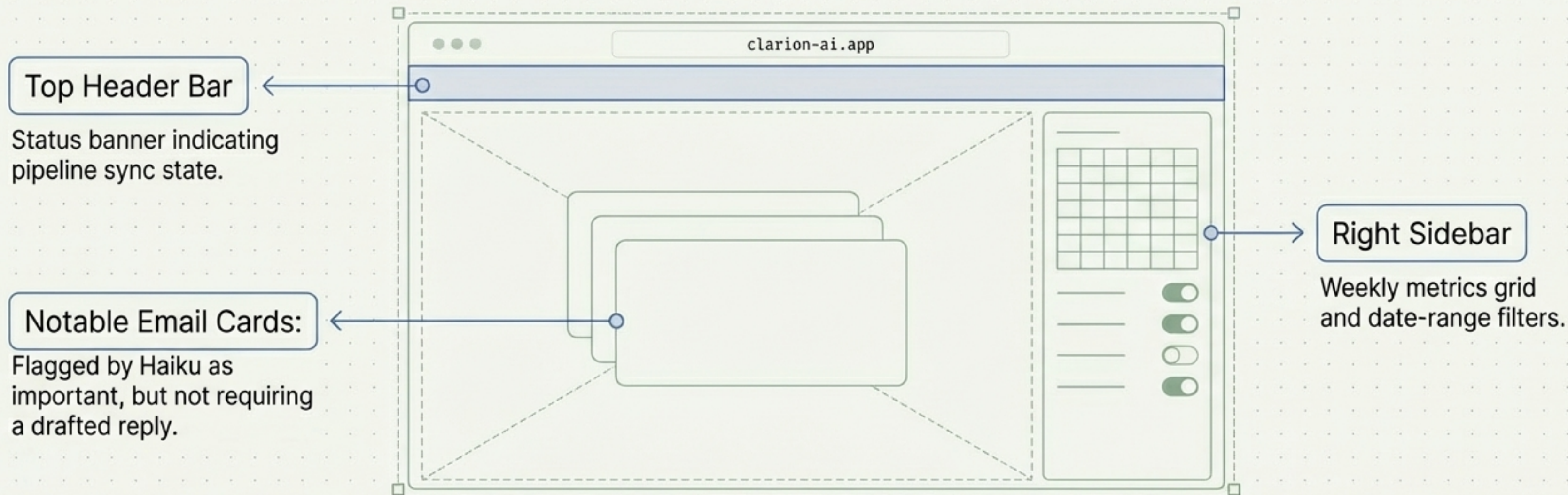
Executes OWA DeleteItem
(HardDelete) in Outlook.

The AI steps aside effortlessly when manual control is assumed. Stale drafts are invisibly swept away, preventing duplicate replies.

Strict Gating Protects Compute and Infrastructure.



The Web Dashboard Closes the Feedback Loop.



User corrections submitted here accumulate until they trigger `check_retrain_needed()`, continuously refining the model output.

The Clarion AI Master Pipeline.

